

Description

The A2 type pressure regulators are compact, direct diaphragm operated regulators are for use in systems where a small inlet or outlet pressure regulator is needed.

Materials

Body Gray iron
Seat Chrome plated
Diaphragm Stainless steel preformed

Specifications

Maximum Fluid Temperature 105°C (220°F)
Minimum Fluid Temperature -45°C (-50°F)
Design Pressure (MRP) 27.6 bar (400 PSIG)

A2A/A2B



A2CK

**Application and Selection Guide**

Type	Function	Operation	Typical Applications
A2B	Control inlet pressure	Regulate at preset inlet pressure Field adjustable	1. Small capacity back pressure regulator 2. Small capacity defrost relief regulator
A2A	Inlet regulator, greater capacity	Open on rise in inlet pressure	
A2BO	Control outlet pressure	Regulate at preset outlet pressure Field adjustable Open on drop in outlet pressure	1. Gas pressure reducing regulator 2. Liquid or oil pressure reducing regulator
A2CK①	Relief regulator.	Regulate inlet pressure. Factory set.	1. Prevent hydrostatic pressure build-up in isolated sections of liquid lines.

① The design of the A2CK valve allows for higher flow and a "quick release" feature. However, it does not offer the same type of controlled response which may be required for a standard regulator application. For those applications where a small regulator is needed for accurate upstream pressure control, an A2A or A2B direct operated regulator should be considered.

Ordering Guide

Type	Description	Ranges Available (Std. is Bold)	Connections Available ③ Style: FPT, SW, WN	Flow④ Coefficient		Net Weight			
				Kv	Cv	Less Strainer		With Strainer	
						kg	lb	kg	lb
A2B	Small Capacity Back Pressure Regulator	V, A, D	¼", ⅜", ½", ¾"	0.43	0.5	3.6	8	4.5	10
A2BK①	Relief, Give pressure setting	A, D		0.34	0.4	3.6	8	4.5	10
A2BP	Differential Regulator for ext. connection	A, D		0.43	0.5	3.6	8	4.5	10
A2BM	Electrically Compensated	V, A, D							
A2BT	Temperature Operated Regulator	1, 2	¼", ⅜", ½", ¾"						
A2A	Small Capacity Back Pressure Regulator	A, D	¼", ⅜", ½", ¾"	1.28	1.5	3.6	8	4.5	10
A2BO1②	Small Capacity Outlet Pressure Regulator	V, D	¼", ⅜", ½", ¾"	0.09	0.1	3.6	8	4.5	10
A2BO2②				0.17	0.2	3.6	8	4.5	10
A2BO4②				0.43	0.5	3.6	8	4.5	10
A2CK	Relief Regulator – Liquid Lines	D	¼", ⅜", ½", ¾"	0.97	1.14	3.6	8	4.5	10

① The A2BK Reseating Relief Regulator can be used for various relief to low side applications including cold liquid line sections where buildup of dangerous hydrostatic pressure is possible. See "Safe Operation" on the back cover.

② Specify for external pressure connection.

③ Bold face type is standard size. FPT will be furnished unless specified otherwise.

④ Flow coefficients are for standard range.
Ranges V = 500mm Hg to 8.3 bar (20 in Hg to 120 psig);
A = 0 to 10.3 bar (0 to 150 psig); D = 5.2 to 19.3 bar (75 to 280 psig)

To Order

Specify: Type, Range, Flange Size, Flange Style, if with Strainer, Gauge and Gauge Valve.

Example: A2B, range A, ¾" WN, with strainer, Q116 gauge, V12 gauge valve.

Capacities (KW)

	Type	ΔP, bar	-40°C	-30°C	-20°C	-10°C	0°C
R22	A2B	0.14	0.42	0.53	0.67	0.84	1.0
	A2A		1.3	1.6	2.0	2.5	3.0
R717	A2B	0.14	1.1	1.5	1.9	2.4	2.9
	A2A		3.3	4.6	5.6	7.3	8.9
	A2B	0.34	1.8	2.3	3.0	3.8	4.7
	A2A		5.3	7.3	8.8	11	14
	A2B	0.70	2.5	3.3	4.2	5.4	6.7
	A2A		7.4	10	12	16	20

Capacities (US Tons)

	Type	ΔP, psi	-40°F	-20°F	0°F	+20°F	+40°F
R22	A2B	2	0.12	0.16	0.20	0.26	0.32
	A2A		0.36	0.47	0.61	0.77	0.96
R717	A2B	2	0.32	0.44	0.58	0.74	0.93
	A2A		0.95	1.3	1.7	2.2	2.8
	A2B	5	0.50	0.67	0.89	1.2	1.4
	A2A		1.5	2.0	2.7	3.4	4.3
	A2B	10	0.71	0.98	1.2	1.6	2.0
	A2A		2.1	2.9	3.7	4.7	6.0